Econometric Game - Case II
(Group 9)

April 9, 2026

Abstract

We investigate the rise of renewables and the increased interconnectivity of electricity grids in Denmark and neighbouring countries. These developments have significant implications for electricity price forecasting, increasing the complexity of the problem. We use a LEAR model to forecast day-ahead prices for 2020 and 2024, extending the model to include cross-zone predictors as well as interactions to account for these structural changes. Our key finding is that prediction performance decreases between 2020 and 2024. We interpret this as evidence that increases in renewables and interconnectivity have decreased forecasting accuracy. We also find that the predictive power of these variables has increased over time.

1 Introduction

Forecasting electricity prices accurately is a crucial problem for a wide range of electricity market participants including traders, producers, grid planners, consumers, and governments and regulators. However, it is also a uniquely complicated issue, due to electricity being non-storable and the need for constant balance between supply and demand, leading to high volatility, seasonality, and periodicity in the price. A key target for forecasters is the day-ahead hourly prices, and many approaches have been proposed for predicting these given the complexity and importance of the forecasting problem. Furthermore, structural changes including increases in renewable energy and in cross-zone transmissions have altered the nature of the prediction problem.

The rise of renewables, particularly wind and solar, is a key recent development in the electricity market. Denmark is particularly reliant on renewable energy: around 50 percent of Danish electricity is generated by wind and solar. Denmark has also seen several expansions in renewable generation in recent years, including the largest wind farm in Scandinavia, Kriegers Flak, which became operational in 2021. Increases in renewable energy have implications for electricity price forecasting: renewables have close to zero marginal cost, leading to prices that are close to zero or even negative, and exhibit high variability, increasing the volatility of electricity prices (Ketterer, 2014; Ziel & Weron, 2018).

Another notable change is the increasing interconnectivity of electricity grids. Denmark is a strong example of this trend: the COBRAcable connecting Denmark to the Netherlands opened in 2019, while the Viking Link connecting Denmark to the United Kingdom (UK) became operational on 2023. These transmission links allow electricity to flow freely in response to supply and demand shocks. Increased cross-zone flows have added another layer of complexity to the forecasting problem, as electricity prices in one zone depend not only on dynamics in that zons but also in interconnected zones (De Blauwe et al., 2025). Further, interconnected zones may have additional connections to other zones, meaning that prices are affected not only by neighbouring zones but also zones further away, creating complex network effects.

In this paper, we address three research questions. Firstly, how can we best forecast step-ahead hourly spot prices for 2024 and 2020. Secondly, how did renewables evolve over the period 2018 to 2024, and how did this impact prices and price forecasting. Thirdly, what spillovers exist between price zones, how did they evolve over the time period, and how did these affect prices and price forecasting. We focus on the DK1 price zone between 2018 and 2024. This is a particularly interesting setting to study these research questions due to

the high penetration of renewables, as well as the large cross-border flows to DK2, Sweden, Germany, Norway, and the UK.

To address our first research question, we employ two different approaches to forecast electricity prices. First, we use a Lasso-Estimated AutoRegressive (LEAR), a statistical model that incorporates past prices as well as a rich set of exogenous variables (Uniejewski et al., 2016). We extend this model to include cross-region predictors to explicitly account for grid interconnectivity. Second, we use a LightGBM model as a machine-learning based benchmark model. We evaluate the approaches against 2024 and 2020 realizations using the root mean squared error (RMSE).

We employ a combination of descriptives and LEAR model outputs to address our second and third research questions. For our second research question, we provide descriptives on the development of renewable energy across zones over time. We also investigate differences in the predictive power of renewables in our LEAR model predictions for 2020 and 2024, and compare the variance of LEAR forecast errors to better understand the impact of renewable expansion on prices and price forecasting. To answer our third research question, we provide descriptives on price zone spillovers and on the evolution of transmission flows between DK1 and connected regions over time. In addition, we compare the performance of the LEAR model with and without cross-zone predictors to investigate their predictive power. Finally, we compare selection frequency of cross-zone

We find XXXXXX

We contribute to the literature by examining

The rest of the paper is structured as follows. Section 2 provides descriptive statistics that address our second and third research questions. In Section 3, we describe our econometric approaches. Section 4 presents our key forecasting results, and . In Section 5, we discuss and interpret our results. Section 6 concludes.

2 Data Description

In this section, we present descriptive results on the production of renewables in DK1 and surrounding zones, as well as empirical patterns showing the evolution of connectivity to neighboring zones. These results motivate our analysis described in Section 3 to investigate the impacts of these shifts on electricity prices and price forecasting.

2.1 Renewables production

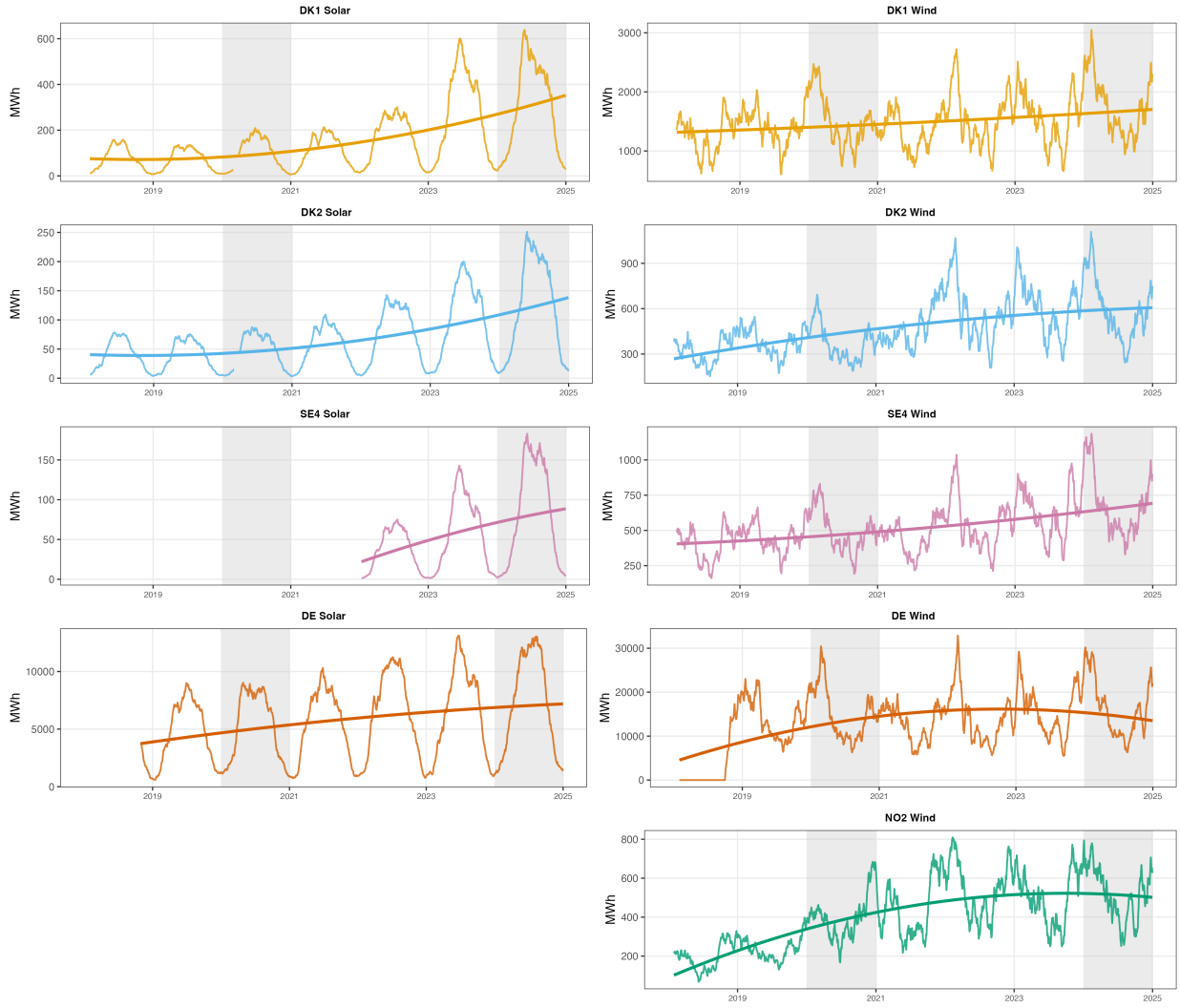


Figure 1: 30-Day Moving Average of Renewable Production by Region (2018–2024)

Notes: Thick lines show quadratic trends. Shaded areas represent 2020 and 2024. Pre-2022 solar data for SE4 and wind data for NO2 are missing.

Figure 1 shows an increase in solar and wind across regions in the period 2018-2024. In DK1, solar production increased at a growing rate, while wind steadily increased over time. We see a very similar trend for solar in DK2, while wind increased more rapidly at the start of the period. We also note that renewable production was consistently lower in DK2 than in DK1 across the sample period. SE4 shows a strong increase in solar and a steady increase in wind, similar to patterns in SK1. In DE, solar increases steadily, while wind increases from zero at the beginning of the sample period. In NO2, we see a strong increase in wind at the beginning of the sample period. Our key takeaway from this graph is that

renewables have increased over the period, not only in DK1 but across all interconnected zones. This indicates the need to consider how prices and forecasting are affected by the rise of renewables, not only in DK1 but also in neighboring zones.

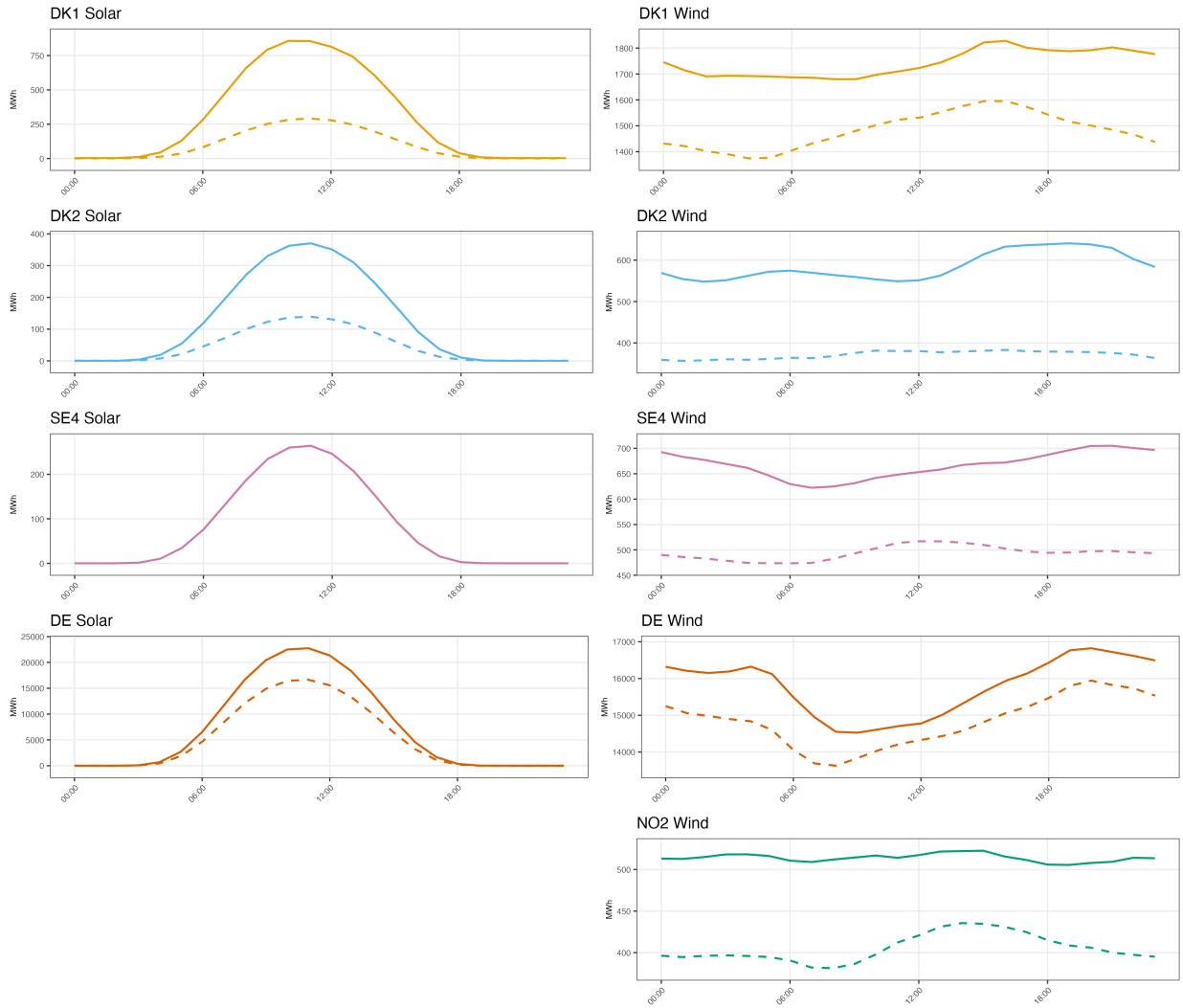


Figure 2: Intraday Variation in Production of Renewables by Region (2020 and 2024)

Notes: Values show yearly averages of production per hour of the day. Solid lines represent 2024, dashed lines 2020. Pre-2022 solar data for SE4 and wind data for NO2 are missing.

Figure 2 depicts the intraday variation in renewable production across regions in 2020 and 2024. In all zones, renewable generation has increased at all times of day, except for solar at night. Interestingly, we see somewhat different hourly patterns for wind generation across zones: in DE there is a stronger dip during the morning. This may reflect differences in the composition of onshore and offshore wind production between zones. This figure illustrates that while renewable production has increased, it is also highly variable across the day. We

expect the rise of renewables to therefore increase the volatility of prices, which may be exacerbated by interconnectivity to regions with highly variable renewables generation.

2.2 Networks in neighboring zones

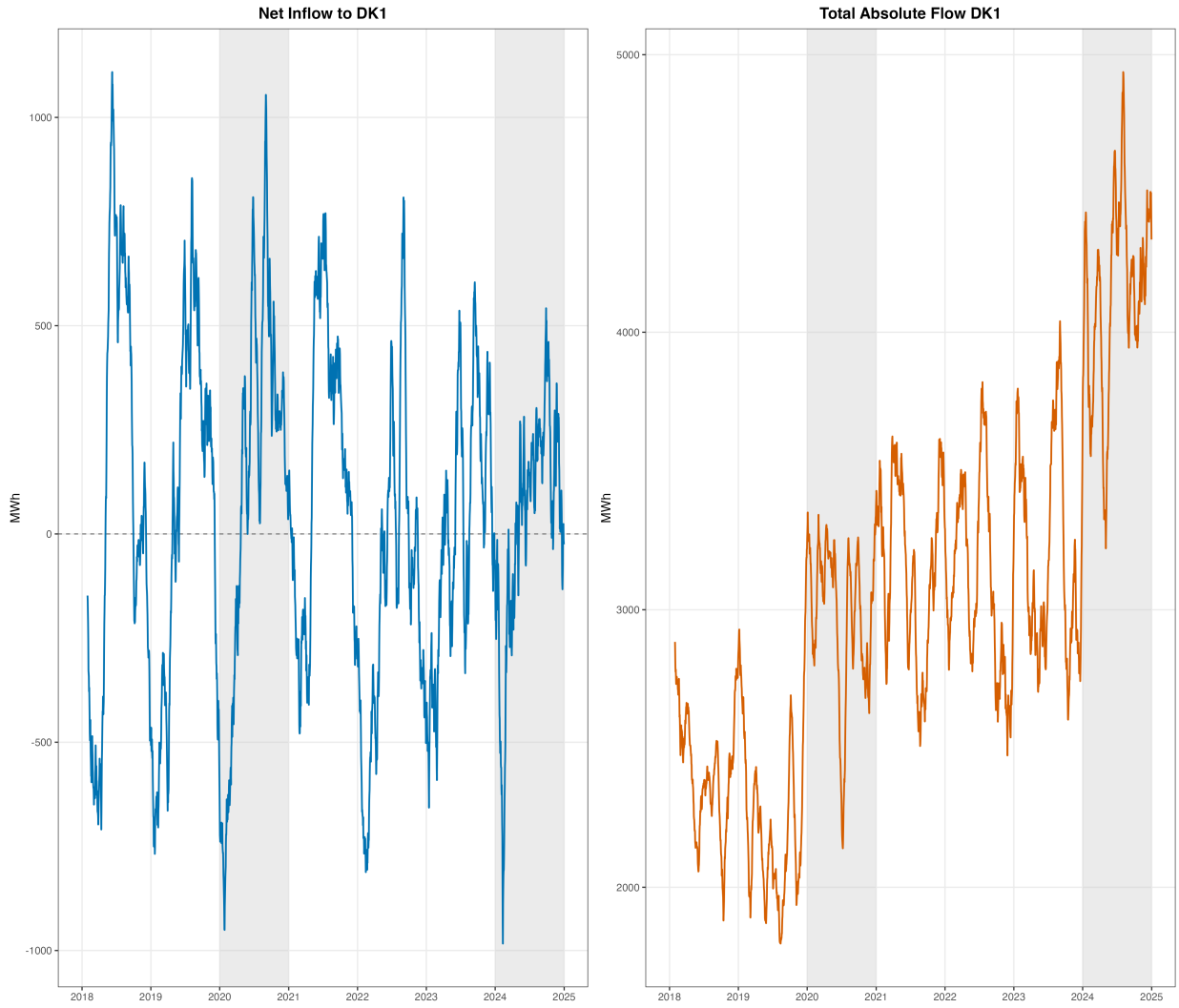


Figure 3: 30-Day Moving Average of Electricity Flows in DK1 (2018–2024)

Notes: Values show 30 day moving averages of Production per hour of the day.

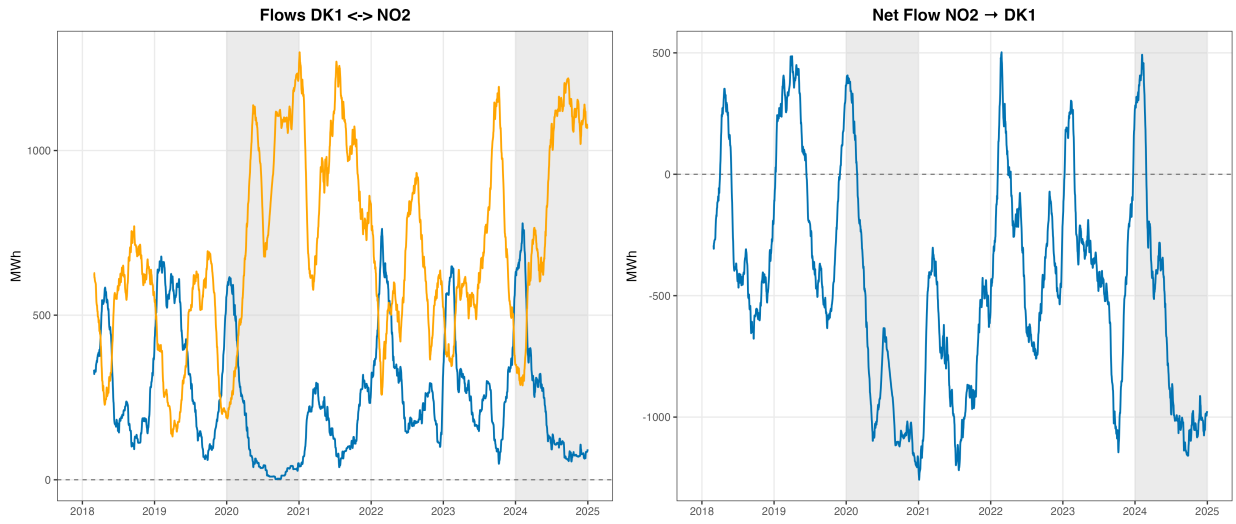


Figure 4: 60-Day Moving Average of Electricity Flows between DK1 and NO2 (2018–2024)

Notes: Left: Yellow are flows from NO2 to DK1 and blue vice-versa. Shaded areas represent 2020 and 2024.

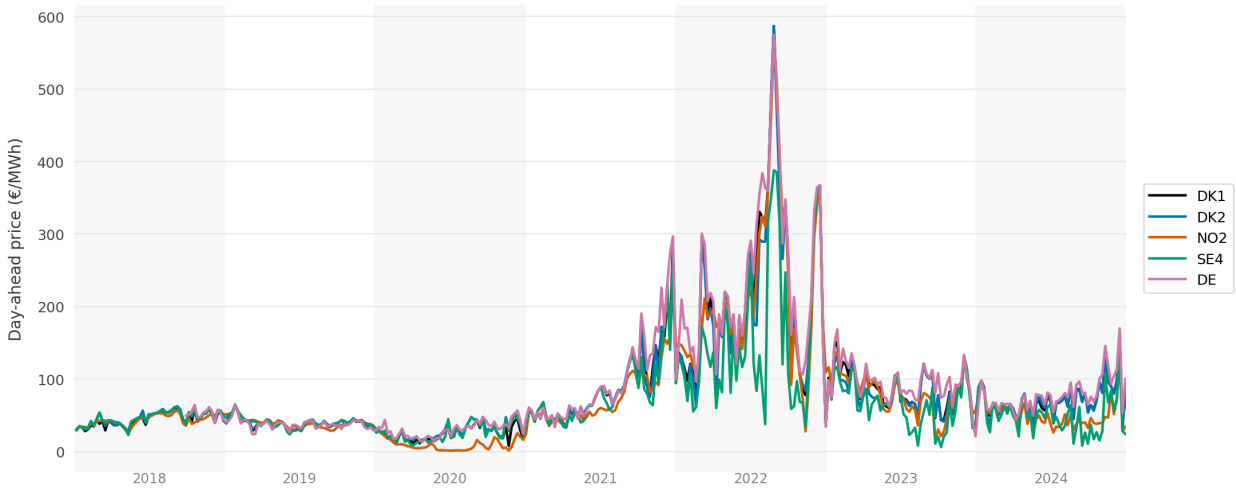


Figure 5: Day-ahead Electricity Prices Across Nordic and German Markets (2018–2024)

Notes: The figure plots weekly average day-ahead spot prices in €/MWh for DK1, DK2, NO2, SE4, and DE over the period 2018–2024. Weekly averages are computed from hourly observations. Source: ENTSO-E Transparency Platform.

Figure 5 shows that prices across zones move largely in tandem, with some differences. The price in DE is typically higher over the sample period than in other zones, while the SE4 price is typically lower. All zones are affected by the energy crisis, although not equally: SE4 exhibits the smallest spike. We also see a significant dip in prices in NO1 in 2020 - this is likely due to an unusual hydropower surplus Norway experienced in 2020. At the end of 2024, we also observe some decoupling of SE4 and NO2 prices from DK1, DK2, and

DE. Overall, this figure underlines that prices across zones are generally strongly correlated across the sample period, reflecting a high degree of grid interconnectivity.

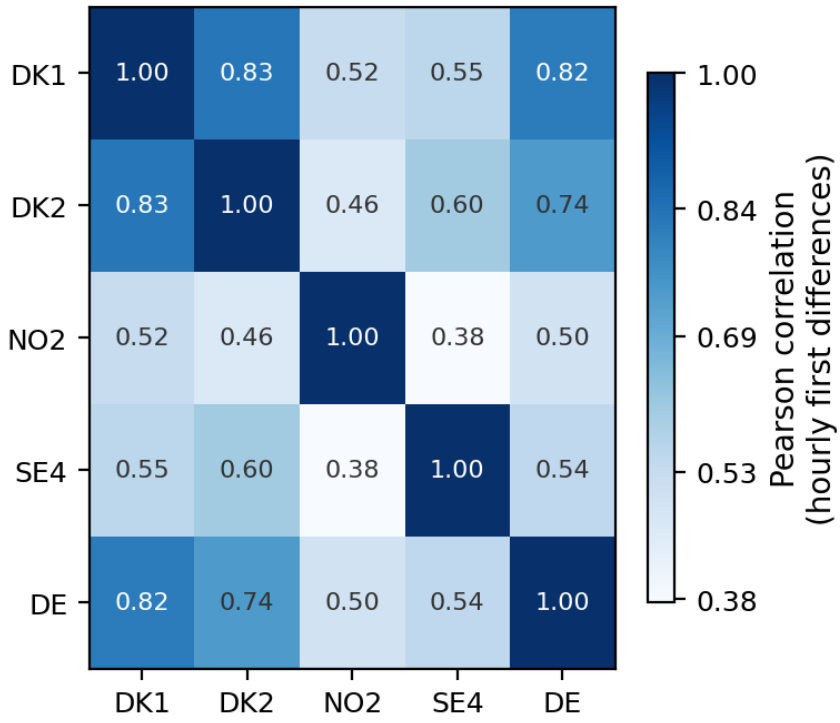


Figure 6: Cross-country Electricity Price Correlations with DK1, Heatmap

Notes: The heatmap reports pooled Pearson correlations of hourly price first differences ($\Delta P_t = P_t - P_{t-1}$) across all hours in 2018–2024. First differencing removes the common price-level trend (in particular the 2021–2022 energy crisis) which would otherwise increase correlations spuriously.

3 Methodology

3.1 LEAR Model

The Lasso-Estimated AutoRegressive (LEAR) model of Lago et al. (2021), originally introduced as *LassoX* by Uniejewski et al. (2016), is a benchmark in day-ahead electricity price forecasting. We produce rolling day-ahead forecasts for all hours of 366 days of 2020 and 2024, extending the model with cross-zone variables.

We train two separate models, one for each forecast year, using a fixed 2-year rolling window. For 2020, the window starts from 2018; for 2024, from 2022. The model is re-estimated daily. As each new day is observed, it enters the training window and the oldest day is dropped, so the forecast for day d always uses the most recent 2 years of realized data

available up to day $d-1$.

Baseline LEAR Specification

Following the benchmark LEAR model (Lago et al., 2021), for each day d, we estimate a separate model for each hour $h \in \{1, \ldots, 24\}$, yielding 24 independently estimated LASSO regressions. For hour h on day d:

For the day-ahead forecast, LEAR estimates a separate model for each hour $h \in \{1, \ldots, 24\}$, yielding 24 independently estimated LASSO regressions:

$$
\begin{aligned}
p_{d,h} = & \underbrace{\sum_{i=1}^{24} \theta_{h,i}\, p_{d-1,i} + \sum_{i=1}^{24} \theta_{h,24+i}\, p_{d-2,i} + \sum_{i=1}^{24} \theta_{h,48+i}\, p_{d-3,i} + \sum_{i=1}^{24} \theta_{h,72+i}\, p_{d-7,i}}_{\text{96 autoregressive terms}} \\
& + \underbrace{\sum_{i=1}^{24} \theta_{h,96+i}\, x^1_{d,i} + \sum_{j=1}^{3}\sum_{i=1}^{24} \theta_{h,120+i}\, x^2_{j,d,i} + \sum_{i=1}^{24} \theta_{h,144+i}\, x^1_{d-1,i} + \sum_{j=1}^{3}\sum_{i=1}^{24} \theta_{h,168+i}\, x^2_{j,d-1,i}}_{\text{192 exogenous terms: days } d \text{ and } d-1} \\
& + \underbrace{\sum_{i=1}^{24} \theta_{h,192+i}\, x^1_{d-7,i} + \sum_{j=1}^{3}\sum_{i=1}^{24} \theta_{h,216+i}\, x^2_{j,d-7,i}}_{\text{96 exogenous terms: day } d-7} + \underbrace{\sum_{i=1}^{7} \theta_{h,240+i}\, z_{d,i}}_{\text{7 day-of-week dummies}} + \underbrace{\phi_h\, b_d}_{\text{bank holiday}} + \varepsilon_{d,h} \qquad (1)
\end{aligned}
$$

Following (Lago et al., 2021), we add three blocks of regressors in the model:

- **Autoregressive terms (96 variables):** All 24 hourly prices from lag days $d-1$, $d-2$, $d-3$, and $d-7$ are included. Lags d-4 to d-6 are excluded as they fall on different weekdays and have low predictive power.
- **Exogenous terms (288 variables):** Four exogenous variables are included at all 24 hours for days d, $d-1$, and $d-7$, yielding $4 \times 3 \times 24 = 288$ regressors. x^1 is the day-ahead total load forecast, providing information on the demand-side. x^2_j for $j \in \{\text{onshore wind, offshore wind, solar}\}$ are the three separate day-ahead renewable generation forecasts, replacing the combined renewable variable of Lago et al. (2021) to distinguish the price effect of each source (we showed this to improve predictive performance in Case A). Higher renewable generation decreases prices.[1]
- **Day-of-week dummies (7 variables):** A binary vector $\mathbf{z}_d = [z_{d,1}, \ldots, z_{d,7}]^\top$ for each day of the week (Lago et al., 2021).

[1]For day d, day-ahead forecasts are used as realized values are unavailable at auction time.

- **Bank holidays (1 variable):** We add a Danish bank holiday dummy, which we showed to improve predictive performance in Case A. Public holidays exhibit electricity demand similar to at weekends, which day-of-week dummies alone cannot capture.

LASSO Estimation

Given the large number of regressors, OLS would suffer from overfitting. Following Lago et al. (2021), we therefore estimate parameters via LASSO (Tibshirani, 1996):

$$\hat{\boldsymbol{\theta}}_h = \arg\min_{\boldsymbol{\theta}_h} \left\{ \sum_{d=8}^{N_d} \left(p_{d,h} - \hat{p}_{d,h}\right)^2 + \lambda \sum_{i=1}^{247} |\theta_{h,i}| \right\} \tag{2}$$

The ℓ_1 penalty shrinks coefficients of irrelevant regressors to exactly zero i.e., performing automatic variable selection among the 247 candidate predictors. Following Lago et al. (2021), λ is selected daily via LARS with in-sample AIC and the model is re-estimated at the optimal λ using coordinate descent.

Variance-Stabilizing Transformation

Electricity prices exhibit heavy tails and frequent spikes. Following Lago et al. (2021), all variables are transformed prior to estimation using the area hyperbolic sine (asinh) transformation:

$$\operatorname{asinh}(x) = \log\left(x + \sqrt{x^2 + 1}\right) \tag{3}$$

where x is the price standardized by the in-sample median and median absolute deviation. Unlike the log transformation, asinh handles zero and negative prices, which occur frequently in our data. Predictions are back-transformed to EUR/MWh after forecasting.

3.2 Extending Lear Using Cross-Zone Predictors

We extend the baseline LEAR with three groups of additional regressors using the cross-zone data. All additional variables enter at lags $d-1$ and $d-7$, following the baseline lag structure of the exogenous variables.

- **Cross-zone prices:** We add day-ahead prices from neighboring zones DK2, DE, NO2, and SE4 at lags $d-1$ and $d-7$.
- **Cross-zone renewable forecasts:** We add day-ahead renewable generation forecasts from neighboring zones DK2, DE, NO2, and SE4 at days d, $d-1$, and $d-7$. We expect that high renewable output in neighboring zones decreases electricity prices in DK1.

- **Transmission flows:** We add net cross-border transmission flows for all corridors between all zones i.e., DK1, DK2, DE, NO2, and SE4 at lags $d-1$ and $d-7$. These variables capture the channels through which cross-zone supply and demand spillovers propagate.

3.3 Extending LEAR using interaction effects

The standard LEAR specification is a linear model: it can capture the level of net load or transmission flows, but not the convex price responses that arise when residual demand is very high or very low, nor the coupling effects that emerge when system stress is simultaneously elevated in neighbouring bidding zones. To address these limitations, we augment the baseline feature vector with a set of economically motivated interaction and nonlinear terms.

Net load nonlinearity. Because the merit-order curve is convex, the marginal price of electricity is a nonlinear function of residual demand. For the Danish DK1 zone we define hourly net load as

$$\widetilde{L}_{d,h} = \widehat{\text{Load}}_{d,h} - \widehat{\text{Offshore}}_{d,h} - \widehat{\text{Onshore}}_{d,h} - \widehat{\text{Solar}}_{d,h}, \tag{4}$$

where all quantities are day-ahead forecasts for delivery hour h on day d. We include both $\widetilde{L}_{d,h}$ and its element-wise square $\widetilde{L}^2_{d,h}$ as additional regressors, allowing the LASSO to fit a piecewise-linear approximation to the curved merit-order relationship.

Cross-border system strain. DK1 is electrically coupled to Germany (DE) and East Denmark (DK2) via high-capacity interconnectors. When residual demand is high in multiple zones simultaneously, cross-border flows become constrained and prices converge upward across the region. We capture this joint-scarcity effect through the interaction terms $\widetilde{L}^{\text{DK1}}_{d,h} \times \widetilde{L}^{\text{DE}}_{d,h}$ and $\widetilde{L}^{\text{DK1}}_{d,h} \times \widetilde{L}^{\text{DK2}}_{d,h}$, where net loads for DE and DK2 are computed analogously using combined wind and solar forecasts available for those zones.

Hydro availability proxy. Norwegian hydropower constitutes the dominant flexible resource accessible to DK1 via the Skagerrak interconnector. When Norwegian reservoirs are well-replenished, cheap hydro imports dampen Danish spot prices; when they are depleted, the dampening effect disappears. Since direct reservoir-level data are not publicly available at the required frequency, we construct a proxy by interacting the 14-day trailing average of observed daily precipitation in the NO2 bidding zone, $\bar{r}^{\text{NO2}}_{d-1}$, with the realised hourly physical flow on the DK1–NO2 interconnector lagged one day, $\phi^{\text{NO2}}_{d-1,h}$. The resulting term $\bar{r}^{\text{NO2}}_{d-1} \cdot \phi^{\text{NO2}}_{d-1,h}$

is positive and large when abundant precipitation coincides with active Norwegian exports to DK1, jointly signalling cheap hydro availability.

Temperature comfort-zone deviation. Electricity demand in Denmark exhibits a U-shaped relationship with temperature: both cold spells and heat waves raise consumption through heating and cooling loads, respectively. Because the LASSO is restricted to linear functions of its inputs, this nonlinearity must be made explicit. We introduce the term $(T_{d-1} - T^*)^2$, where T_{d-1} is the daily mean temperature observed in DK1 on the previous day and $T^* = 18\,^\circ\mathrm{C}$ is an approximate thermal comfort threshold. The squared deviation is identical for symmetric departures above and below T^*, enabling the model to recognise that both extremes are associated with elevated demand and upward price pressure.

In total, the interaction extension adds 121 regressors to the baseline feature vector: 24 hourly values each for $\widetilde{L}_{d,h}$, $\widetilde{L}^2_{d,h}$, the DK1×DE coupling, and the DK1×DK2 coupling, plus 24 values for the hydro proxy and a single scalar for the temperature term. All regressors enter the LASSO on equal footing alongside the baseline predictors, and the penalty-parameter selection and asinh variance-stabilising transformation are applied identically to the augmented feature matrix.

3.4 Comparison of 2020 and 2024 Forecasts

Predictive power of renewables: We track LASSO selection of variables and magnitudes of the coefficients on each renewable variable, and compare these for 2020 and 2024. Renewable capacity expanded substantially between 2020 and 2024 as shown in Section 2, so we expect LASSO to give more weight to renewables in 2024 and larger coefficient magnitudes.

Impact on price dynamics: To assess whether renewable expansion affected price volatility, we compare the variance of LEAR forecast errors across 2020 and 2024. Higher residual variance in 2024 would indicate that increased renewable production decreased price predictability. For example, with renewable expansion, high wind output can drive prices to zero or even negative, while sudden drops in wind production cause sharp upward spikes.

3.5 Comparison of the Baseline and Cross-Zone Extended Model

Spillovers across price zones: To assess spillovers on prices across different price zones, we document LASSO selection frequency of cross-zone price variables i.e., prices in DK2,

DE, NO2, SE4. A variable consistently selected by LASSO indicates high predictive power of prices in that zone, implying price spillovers through the transmission network.

Predictive power of cross-zone model: To quantify whether cross-zone variables improve prediction, we compare RMSE between the baseline LEAR and the extended LEAR model with cross-zone predictors. An improvement in RMSE indicates that cross-zone variables increase the predictive power for electricity prices in DK1.

Change in network structure over time. To assess whether the network structure around Denmark changed in its impact on DK1 electricity prices between 2020 and 2024, we compare LASSO selection frequency of cross-zone variables in 2020 and 2024. Higher selection frequency in 2024 relative to 2020 would indicate that the impact of the network structure around Denmark on electricity prices increased over time.

3.6 LightGBM Benchmark

As an additional benchmark, we estimate a gradient-boosted tree model using LightGBM. Unlike the LEAR model, which fits 24 separate hour-specific LASSO regressions, we train a single global model with `hour` included as a categorical feature, allowing the model to capture hour-specific price dynamics through tree interactions. The features follow the full LEAR specification: autoregressive price lags for all bidding zones at $d-1$, $d-2$, $d-3$, and $d-7$; load, offshore, onshore and solar production forecasts for all bidding zones at lags d, $d-1$, and $d-7$; energy commodity prices (coal, EUA, natural gas, crude oil) at lags $d-1$ and $d-7$; macrofinancial variables (EuroStoxx 50, MSCI EU, USD/EUR, VIX) at lags $d-1$ and $d-7$; per-channel cross-border transmission net flows at lags $d-1$ and $d-7$; the five interaction terms described above; and calendar dummies (hour, day-of-week, month, day-of-year, holiday, summer break, winter break).

To account for the distinct market regimes across the forecasting horizons, hyperparameter tuning is conducted in two separate phases. For the 2020 prediction task, parameters are optimized using data from 2018 to 2019, utilizing a chronological time-series split (training on 2018, validating on 2019). Similarly, for the 2024 prediction task, tuning is performed on data from 2022 to 2023 (training on 2022, validating on 2023). In both phases, hyperparameters are selected via Bayesian optimization using Optuna with the Tree-structured Parzen Estimator (TPE) sampler, minimizing the RMSE on the respective validation set. To ensure computational efficiency within local hardware constraints, the search space fo-

cuses on the most impactful structural parameters: number of leaves, maximum tree depth, minimum child samples, and learning rate, implemented alongside early stopping. During the out-of-sample prediction phases for both 2020 and 2024, the optimal hyperparameters are held constant for their respective years. The forecasts are generated using a fixed rolling calibration window (e.g., 365 days), where the final model is dynamically re-estimated every day to capture shifting short-term market dynamics before generating the subsequent day-ahead hourly forecasts.

4 Emprical Results

5 Discussion and Economic Interpretation

6 Conclusion

Electricity price forecasting is a complex problem which has received considerable attention in the literature, and spawned a variety of approaches. In this paper, we compare three key approaches, the structural, LEAR models, and a popular ML framework, to forecast electricity prices for 2024, using only data from 2018-2023. We study two horizons.

We first document a strong negative relationship between renewables and prices, and a strong positive relationship between load and prices. We also document high autocorrelation of prices, and high volatility, periodicity, and seasonality of key variables.

We next investigate how the different approaches compare in forecasting prices over short- and long-term horizons. In line with our first hypothesis, the LEAR model performs better in the short-term, although the LightGBM still has the upper hand on the immediate day-ahead forecast. This likely reflects that in the short-term, autocorrelation plays an important role, which is better captured in LEAR than in our LightGBM architecture. In line with our second hypothesis, the structural model does perform better in the longer term with its economic grounding.

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Statement on hardware and software

All analysis was conducted using Python and R. Overleaf was used to write the report. Standard laptop computing was used, no cloud resources were used.

Statement on use of AI tools

Claude was used to assist with Python and R coding (include code snippet generation and debugging), and clarification of econometric concepts. ChatGPT was used to generate code snippets for data manipulation and model estimation. All writing, interpretation of results, and model specification decisions in this report are the authors' own work. Claude and ChatGPT were not used to generate the written report, explanations, or interpretation.